

Algebra I Polynomials Intro Cheatsheet

Module 8

Vocabulary

monomial: one term, binomial: two terms, trinomial: three terms

Degree of a term:

$$3x^4 \Rightarrow \text{degree } 4$$

Degree of a polynomial is the greatest term degree.

Adding and Subtracting

Combine like terms.

$$(3x^2 + 5x - 1) + (2x^2 - x + 7) = 5x^2 + 4x + 6$$

Distribute the negative when subtracting:

$$(4x^2 - 3x) - (x^2 + 6x - 2) = 3x^2 - 9x + 2$$

Multiplying

Monomial times polynomial:

$$a(b + c + d) = ab + ac + ad$$

Binomial product:

$$(a + b)(c + d) = ac + ad + bc + bd$$

Special Products

$$(a + b)^2 = a^2 + 2ab + b^2$$

$$(a - b)^2 = a^2 - 2ab + b^2$$

$$(a + b)(a - b) = a^2 - b^2$$

Factoring GCF

Find the greatest common factor of all terms.

$$6x^3 + 9x^2 = 3x^2(2x + 3)$$

Factoring Trinomials

For

$$x^2 + bx + c$$

find two numbers that multiply to c and add to b .

$$x^2 + 7x + 12 = (x + 3)(x + 4)$$

Factoring by Grouping

$$\begin{aligned} & ax + ay + bx + by \\ &= a(x + y) + b(x + y) \\ &= (a + b)(x + y) \end{aligned}$$

Quick Checks

- Put polynomials in descending degree order.
- Factor out the GCF first.
- Check factoring by multiplying back.
- Difference of squares needs subtraction between two square terms.